

RISK ASSESSMENT & METHOD STATEMENT (RAMS)

Activity: Bored Cast-in-Situ Pile Installation

Project: Al-Noor Hospital Expansion | Ref: RAMS-AHX-PIL-002 Rev 2 | Date: 10 Feb 2024

SECTION 1 — SCOPE OF WORK

This RAMS covers the installation of 48 No. bored cast-in-situ piles, diameters ranging from 600mm to 900mm and lengths 18.0m to 25.0m, using the BAUER BG-28 drilling rig. Works include: establishment of pile positions, rig set-up, bore installation with temporary casing and bentonite slurry stabilisation, fabrication and installation of reinforcement cages, and concrete placement by tremie pipe method.

SECTION 2 — HAZARD IDENTIFICATION & RISK ASSESSMENT

Hazard	Persons at Risk	Likelihood (1-5)	Severity (1-5)	Risk Rating	Control Measures	Residual Risk
Rig overturning / collapse	All site personnel	2	5	HIGH (10)	Daily rig inspection; max ground bearing pressure checked; level ground; exclusion zone 1.5x rig height;	LOW (3)
Falling objects from rig	Workers within 15m	5	4	HIGH (12)	Hard hat mandatory in exclusion zone; tool tethering; drop netting; permit to work at height;	LOW (3)
Pile bore collapse / person falling into open pits	Piling crew & supervisors	2	5	HIGH (10)	Permanent steel casing minimum 3m below ground; bore covers placed immediately when drilling is complete;	LOW (2)
Bentonite slurry hazard (splashing/recontact)	Piling crew	4	2	MEDIUM (8)	Safety goggles and chemical resistant gloves mandatory; eye wash station within 10m; CO2 monitoring;	LOW (4)
Concrete placement — handover vibration (HAV)	Concrete pump operator	2	2	MEDIUM (8)	Rotation of workers; max 2 hrs continuous exposure; HAV tool vibration maintained; health surveillance;	LOW (4)
Rebar handling — cuts, punctures, hands struck by	Rebar fixers	5	3	MEDIUM (12)	Cut-resistant gloves; capped rebar ends; mechanical lifting tools; no lifting >25kg; banksman directed;	LOW (3)
Working near underground services	Excavation operators, supervisors	5	5	HIGH (10)	Services survey completed (ref GT-2023-044A); all services marked with paint; hand-dig with care;	LOW (2)
Heat stress (ambient temperature >35°C)	All site personnel	4	3	MEDIUM (12)	Water stations every 50m; rest breaks 15min/hour when >35°C; acclimatisation period for new staff;	LOW (4)

SECTION 3 — MANDATORY PPE REQUIREMENTS

- SAFETY CRITICAL: Hard hat (EN 397) — mandatory for ALL personnel on site at all times.
- SAFETY CRITICAL: Safety footwear steel toe-cap and midsole (EN ISO 20345:2011 S3 SRC) — mandatory for ALL personnel.
- SAFETY CRITICAL: High-visibility vest/jacket Class 3 (EN ISO 20471) — mandatory for ALL personnel.
- Safety goggles (EN 166) — mandatory during concrete and bentonite operations.
- Chemical-resistant gloves (EN 374) — mandatory during bentonite mixing and placement.
- Cut-resistant gloves (EN 388 Level D) — mandatory for all rebar handling operations.
- Safety harness full body (EN 361) with double lanyard — MANDATORY for all work at height above 2.0m.
- RPE (FFP2 dust mask) — mandatory during dry boring operations and concrete break-down.
- Hearing protection (EN 352) — mandatory within 2m of operating piling rig.

SECTION 4 — PERMITS TO WORK REQUIRED

Permit Type	When Required	Issuing Authority	Duration
Permit to Dig	Before any ground-breaking activity	Site Manager	Per shift
Hot Work Permit	Welding of rebar couplers on site	HSE Officer	Per task (max 8 hrs)
Permit to Work at Height	Any work above 2.0m (mast inspection, crane ops)	HSE Officer	Per shift
Confined Space Permit	Inspection of bore exceeding 1.2m depth	HSE Officer + Standby rescuer	Per entry

Crane Lift Permit	All crane lifts over 1 tonne or within 6m of Overhead Cables	Crane Supervisor + Rigger	Per lift
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SECTION 5 — EMERGENCY PROCEDURE

Emergency Assembly Point: Car park, main entrance gate (West side). Site Emergency Number: +971-50-123-4567 (Site Manager Arun Pillai). Nearest Hospital: Al-Noor Hospital Main Campus — 1.2km south of site (irony noted). Abu Dhabi Civil Defence Emergency: 997. UAE Ambulance: 998.

In the event of a serious injury: (1) Do not move the injured person unless in immediate danger; (2) Call 998 immediately; (3) Notify Site Manager; (4) Preserve scene for investigation; (5) Complete incident report within 1 hour.